K L UNIVERSITY

FRESHMAN ENGINEERING DEPARTMENT

A Project Based Lab Report

On

Lost Item Logger System for

College Campus (LIL-CAMP)

SUBMITTED BY:

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UNDER THE ESTEEMED GUIDANCE OF

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CERTIFICATE

This is to certify that the project based laboratory report entitled "**Lost Item Logger System for College Campus (LIL-CAMP)**" submitted by MEKALA SAKETH NAGA SRI NAIDU bearing Regd. No. 2500040200to the **Department of COMPUTER SCIENCE ENGINEERING , KL University** in partial fulfillment of the requirements for the completion of a project based Laboratory in ""**PYTHON PROGRAMMING (25SC2007)**" course in I B Tech // Semester, is a Bonafede record of the work carried out by him under my supervision during the academic year 2025 – 2026.

PROJECT SUPERVISOR

Mr. REPUDI PITCHIAH

HEAD OF THE DEPARTMENT

Dr. D. HARITHA

ACKNOWLEDGEMENTS

It is great pleasure for me to express my gratitude to our honorable President.

Sri. Koneru Satyanarayana, for giving the opportunity and platform with facilities in accomplishing the project-based laboratory report.

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Finally, we are pleased to acknowledge our indebtedness to all those who devoted themselves directly orindirectly to making this project a success.

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ABSTRACT

The Automated Household Inventory & Intelligent Consumption Analytics System (HI-CAS) is a Python-based application designed to automate and optimize household inventory management. Many households rely on manual tracking methods such as memory or handwritten lists, which often lead to over- purchasing, stock shortages, expired items, and unnecessary financial loss. HI- CAS addresses these issues by providing a structured and intelligent system for managing daily household items efficiently.

The system enables users to add, update, delete, and monitor inventory items while tracking quantities and expiry dates. It generates low-stock alerts and expiry notifications to prevent shortages and reduce wastage. In addition, the system performs basic consumption analytics to analyze usage patterns and support better purchasing decisions.

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INTRODUCTION

The **Lost Item Logger System for College Campus (LIL-CAMP)** is a Python-based project developed to address a common problem faced in college environments—misplacement of personal belongings. In busy campuses, students and staff frequently lose important items such as ID cards, books, mobile phones, wallets, keys, and electronic accessories. Recovering these items becomes difficult due to the absence of a centralized system for reporting and tracking lost and found objects.

Traditionally, students rely on manual methods such as asking around, checking notice boards, or contacting administrative offices. These approaches are time-consuming, inefficient, and often unsuccessful due to poor communication and lack of proper record maintenance. There is no systematic way to store information about lost items or match them with found items.

To overcome these challenges, the LIL-CAMP system provides a **digital solution** that enables users to register lost and found items in a structured format. The system stores essential details such as item name, description, location, date, and contact information, making it easier to search and retrieve information. It also lays the foundation for advanced features like automatic matching, analytics, and reporting.

AIM :

To develop a Python-based script to **capture and register basic lost-item details** including:

- Item name
- Description
- Location
- Date lost
- Contact information

This ensures that the system can store records in an organized manner, laying the foundation for tracking and recovery of lost items on campus.

ADVANTAGES

- **Efficiency:** Automates lost-item registration, reducing manual effort.
- **Structured Storage:** Stores item details in an organized manner for easy retrieval.
- **Foundation for Analytics:** Provides the basis for advanced modules like lost-found matching.
- **Modular Design:** Week-1 module follows object-oriented design, making future integration seamless.

DISADVANTAGES

- Manual Entry: Initial setup requires manual data input.
- Limited Functionality: Week-1 module only records items; no matching or alerts yet.
- Dependency on Accurate Input: Incorrect data entry can affect later matching and recovery.

FUTURE IMPLEMENTATION

Implement **automatic lost-found matching** based on item name and description.

Add **search and filtering options** by category, location, or date.

Generate **analytics reports** showing frequently lost items, top locations, and recovery trends.

-> Introduce **persistent storage** using CSV/JSON files.

-> Expand into **GUI or web application** for user-friendly interaction.

SYSTEM REQUIREMENTS

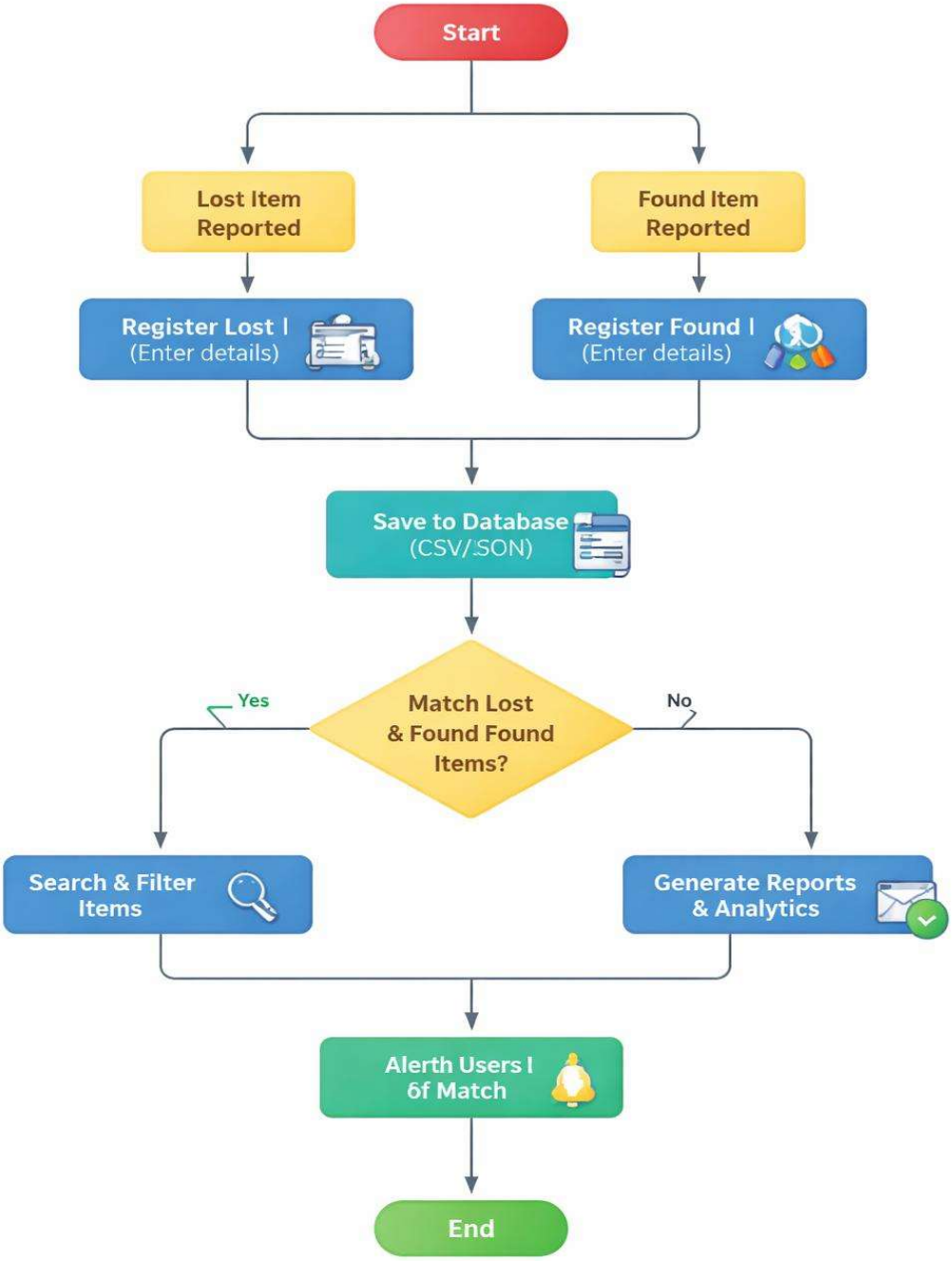
Software Requirements:

- Python 3.x
- IDE: VS Code / PyCharm / IDLE

Hardware Requirements:

- Laptop or PC with minimum 4 GB RAM
- Keyboard, Monitor

FLOW CHART DIAGRAM :



ALGORITHM

Step 1: Start program.

Step 2: Prompt user to enter lost-item details:

- Item name
- Description
- Location
- Date lost
- Contact info

Step 3: Create an **Item object** with these details.

Step 4: Store the object in a list (in-memory).

Step 5: Display the registered item details to the user.

Step 6: End program.

IMPLEMENTATION

```
class Item:
    def __init__(self, name, description, location, date_lost, contact_info):
        self.name = name
        self.description = description
        self.location = location
        self.date_lost = date_lost
        self.contact_info = contact_info

    def __str__(self):
        return f"Item: {self.name}\nDescription: {self.description}\nLocation:
{self.location}\nDate Lost: {self.date_lost}\nContact: {self.contact_info}"

def capture_lost_item():
    print("Enter details for lost item:")
    name = input("Item name: ")
    description = input("Description: ")
    location = input("Location: ")
    date_lost = input("Date Lost (YYYY-MM-DD): ")
    contact_info = input("Contact info: ")

    item = Item(name, description, location, date_lost, contact_info)
    print("\nLost item registered successfully!")
    print(item)

if __name__ == "__main__":
    capture_lost_item()
```

screen short :

```python
class Item:
    def __init__(self, name, description, location, date_lost, contact_info):
        self.name = name
        self.description = description
        self.location = location
        self.date_lost = date_lost
        self.contact_info = contact_info
    def __str__(self):
        return f"Item: {self.name}\nDescription: {self.description}\nLocation: {self
            .location}\nDate Lost: {self.date_lost}\nContact: {self.contact_info}"
def capture_lost_item():
    print("Enter details for lost item:")
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    date_lost = input("Date Lost (YYYY-MM-DD): ")
    contact_info = input("Contact info: ")

    item = Item(name, description, location, date_lost, contact_info)
    print("\nLost item registered successfully!")
    print(item)
if __name__ == "__main__":
    capture_lost_item()
```

OUTPUT:

Output

```
Enter details for lost item:
Item name: id card
Description: green tag
Location: kl hostel
Date Lost (YYYY-MM-DD): 2026-4-09
Contact info: 9876543210

Lost item registered successfully!
Item: id card
Description: green tag
Location: kl hostel
Date Lost: 2026-4-09
Contact: 9876543210

=== Code Execution Successful ===
```

Conclusion :

The **Lost Item Logger System for College Campus (LIL-CAMP)** successfully demonstrates how a real-world problem can be addressed using the concepts of Python programming and computational thinking. The system provides a structured and efficient approach to recording, tracking, and managing lost and found items within a college campus.

Through this project, a digital platform is established that replaces traditional manual methods with an organized system capable of storing item details, improving communication, and increasing the chances of recovery. The modular design ensures that each component of the system—data collection, validation, storage, analysis, and visualization—can be developed and enhanced independently.